

Disproof of Conjecture 00000001196: the largest coefficient of $Q_{(k)}Q_{(k)}$ is not 2^{k-1}

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Abstract

Conjecture 00000001196 concerns $M(\lambda, \mu)$, the largest Littlewood–Richardson coefficient in the product of two Schur Q -functions. It claims $M(\lambda, \mu) \leq \min(2^{\ell(\lambda)}, 2^{\ell(\mu)}, f^\nu)$ and $M((k), (k)) = 2^{k-1}$. We prove that for every $k \geq 1$

$$Q_{(k)}Q_{(k)} = 2 \sum_{j=0}^{k-1} Q_{(2k-j, j)}, \quad P_{(k)}P_{(k)} = P_{(2k)} + 2 \sum_{j=1}^{k-1} P_{(2k-j, j)}.$$

So $M((k), (k)) = 2$ for all $k \geq 2$ in both standard normalisations, not 2^{k-1} . The claim first fails at $k = 3$: $Q_3Q_3 = 2Q_6 + 2Q_{51} + 2Q_{42}$, so $M((3), (3)) = 2 \neq 4$. We treat all eight normalisation readings $(X_{(k)}Y_{(k)})$ expanded in the Z -basis, $X, Y, Z \in \{P, Q\}$. Under each of them, at $\lambda = \mu = (3)$, either the value clause or the bound clause of the conjecture fails. The $k = 3$ case is formalised from scratch in Lean 4 (core library only). Schur Q -functions are built from their generating function and Schur’s Pfaffian, and the expansion coefficients are proved unique. An independent Python script recomputes everything from marked shifted tableaux.

1 The conjecture and the clause we refute

The statement (English version) reads:

Definition: $M(\lambda, \mu)$ is the maximum of the Littlewood–Richardson coefficients $N_{\lambda\mu}^\nu$ of Schur Q -functions at fixed shapes. *Conjecture:* $M(\lambda, \mu) \leq \min(2^{\ell(\lambda)}, 2^{\ell(\mu)}, f^\nu)$, with value exactly 2^{k-1} at $\lambda = \mu = (k)$; the extremum is attained on shifted rectangles.

The Chinese version says the same thing. So $M(\lambda, \mu) = \max_\nu N_{\lambda\mu}^\nu$, where $N_{\lambda\mu}^\nu$ are the structure constants of the product of the Q -functions indexed by λ and μ . The conjecture is a conjunction, and we refute the clause

$$M((k), (k)) = 2^{k-1} \quad \text{for all } k \geq 1. \tag{V}$$

It is stated for every k , so one value of k refutes it. We also use a consequence of the bound clause. The quantity f^ν is not defined in the statement, but whatever it is, $\min(2^{\ell(\lambda)}, 2^{\ell(\mu)}, f^\nu) \leq 2^{\ell(\lambda)}$. So the bound clause implies

$$M((k), (k)) \leq 2^{\ell((k))} = 2. \tag{B}$$

Readings. Two normalisations of Schur Q -functions are in common use: Q_λ and $P_\lambda = 2^{-\ell(\lambda)}Q_\lambda$. The “Littlewood–Richardson coefficients of Schur Q -functions” are, in the standard literature, either Stembridge’s *shifted LR coefficients* $f_{\lambda\mu}^\nu$, defined by $P_\lambda P_\mu = \sum_\nu f_{\lambda\mu}^\nu P_\nu$ [2], or the structure constants of the Q -basis, $Q_\lambda Q_\mu = \sum_\nu 2^{\ell(\lambda)+\ell(\mu)-\ell(\nu)} f_{\lambda\mu}^\nu Q_\nu$ ([1, III.8]). We call these the *standard readings*. To be safe we also consider every mixed convention: the reading (X, Y, Z) with $X, Y, Z \in \{P, Q\}$ expands $X_{(k)}Y_{(k)}$ in the basis $\{Z_\nu\}$ and takes the maximal coefficient. There are 8 readings (6 up to the symmetry $X \leftrightarrow Y$). Coefficients may be rational.

2 Schur Q -functions

We follow Macdonald [1, III.8]. Let $x = (x_1, x_2, \dots)$ and define q_r by

$$\sum_{r \geq 0} q_r t^r = \prod_i \frac{1 + x_i t}{1 - x_i t} = \prod_i \left(1 + \sum_{s \geq 1} 2x_i^s t^s\right).$$

Then $q_0 = 1$, and since $Q(t)Q(-t) = 1$ for $Q(t) = \sum_r q_r t^r$,

$$\sum_{i=0}^r (-1)^i q_i q_{r-i} = 0 \quad (r \geq 1). \quad (1)$$

For $r > s \geq 0$ put

$$Q_{(r,s)} = q_r q_s + 2 \sum_{i=1}^s (-1)^i q_{r+i} q_{s-i}, \quad (2)$$

so $Q_{(r,0)} = Q_{(r)} = q_r$. For a strict partition ν of even length m (append a part 0 if the length is odd), $Q_\nu = \text{Pf}(Q_{(\nu_i, \nu_j)})_{1 \leq i, j \leq m}$. For $\nu = (a, b, c)$ this is

$$Q_{(a,b,c)} = Q_{(a,b)} q_c - Q_{(a,c)} q_b + q_a Q_{(b,c)}. \quad (3)$$

Let $P_\nu = 2^{-\ell(\nu)} Q_\nu$. The Q_ν (ν strict) form a basis of $\Gamma_{\mathbb{Q}} = \mathbb{Q}[q_1, q_2, \dots]$ [1, III (8.9)], so every structure constant is well defined. Equivalently, Q_ν is the generating function of marked shifted tableaux of shape ν [1, III (8.16)]. Our Python check uses that description.

Setting $x_i = 0$ for $i > n$ is a ring homomorphism. It sends q_r to $q_r(x_1, \dots, x_n)$, the coefficient of t^r in $\prod_{i \leq n} (1 + x_i t)/(1 - x_i t)$, and hence sends Q_ν to $Q_\nu(x_1, \dots, x_n)$, computed by the same formulas (2), (3).

3 The product $Q_{(k)}Q_{(k)}$

Theorem 1. *For every $k \geq 1$,*

$$Q_{(k)}Q_{(k)} = 2 \sum_{j=0}^{k-1} Q_{(2k-j,j)}.$$

Proof. Let $S = \sum_{j=0}^{k-1} Q_{(2k-j,j)}$. By (2), S is a sum of terms $q_{2k-j+i} q_{j-i}$ with $0 \leq i \leq j \leq k-1$. The coefficient is 1 for $i = 0$ and $2(-1)^i$ for $i \geq 1$. Fix $v = j - i \in \{0, \dots, k-1\}$. Then i runs over $0, \dots, n$ with $n = k-1-v$, and the total coefficient of $q_{2k-v} q_v$ is $1 + 2 \sum_{i=1}^n (-1)^i = (-1)^n$. Hence

$$S = (-1)^{k-1} \sum_{v=0}^{k-1} (-1)^v q_v q_{2k-v}.$$

In (1) with $r = 2k$, the terms i and $2k - i$ are equal. So $2 \sum_{v=0}^{k-1} (-1)^v q_v q_{2k-v} + (-1)^k q_k^2 = 0$, that is, $\sum_{v=0}^{k-1} (-1)^v q_v q_{2k-v} = (-1)^{k-1} q_k^2 / 2$. Therefore $S = q_k^2 / 2$, and $Q_{(k)} Q_{(k)} = q_k^2 = 2S$. $\square$

Corollary 2. *Let $k \geq 2$. In the bases $\{Q_\nu\}$ and $\{P_\nu\}$ the products are*

$$\begin{aligned} Q_k Q_k &= 2Q_{(2k)} + 2 \sum_{j=1}^{k-1} Q_{(2k-j,j)} = 4P_{(2k)} + 8 \sum_{j=1}^{k-1} P_{(2k-j,j)}, \\ P_k Q_k &= Q_k P_k = Q_{(2k)} + \sum_{j=1}^{k-1} Q_{(2k-j,j)} = 2P_{(2k)} + 4 \sum_{j=1}^{k-1} P_{(2k-j,j)}, \\ P_k P_k &= \frac{1}{2} Q_{(2k)} + \frac{1}{2} \sum_{j=1}^{k-1} Q_{(2k-j,j)} = P_{(2k)} + 2 \sum_{j=1}^{k-1} P_{(2k-j,j)}. \end{aligned}$$

These expansions are unique, so $M((k), (k))$ is as in Table 1.

Proof. Use $P_k = Q_k / 2$, $Q_{(2k)} = 2P_{(2k)}$ and $Q_{(2k-j,j)} = 4P_{(2k-j,j)}$ for $1 \leq j \leq k-1$. Uniqueness holds because $\{Q_\nu\}$ and $\{P_\nu\}$ are bases. $\square$

In particular $f_{(k)(k)}^{(2k)} = 1$ and $f_{(k)(k)}^{(2k-j,j)} = 2$, in agreement with the shifted Pieri rule [1, III (8.15)].

reading	$P \cdot P$ in P	$Q \cdot Q$ in Q	$P \cdot Q$ in P	$P \cdot Q$ in Q	$Q \cdot Q$ in P	$P \cdot P$ in Q
$k = 1$	1	2	2	1	4	1/2
$k \geq 2$	2	2	4	1	8	1/2

Table 1: $M((k), (k))$ under each reading. The conjecture claims 2^{k-1} ; the first two columns are the standard readings. ($Q \cdot P$ is the same as $P \cdot Q$.)

Theorem 3.

1. *In both standard readings, $M((k), (k)) = 2^{k-1}$ holds for $k = 1, 2$ and fails for every $k \geq 3$. The smallest counterexample is $k = 3$: $Q_3 Q_3 = 2Q_6 + 2Q_{51} + 2Q_{42}$ and $P_3 P_3 = P_6 + 2P_{51} + 2P_{42}$, so $M((3), (3)) = 2 \neq 4$.*
2. *Under each of the eight readings, at $\lambda = \mu = (3)$, (V) or (B) fails. Hence the conjecture is false under every reading.*

Proof. (1) In Table 1, the first two columns give $1 = 2^0$ at $k = 1$, $2 = 2^1$ at $k = 2$, and $2 < 2^{k-1}$ for $k \geq 3$. (2) At $k = 3$ the claimed value is 4. The readings $P \cdot P$ in P , $Q \cdot Q$ in Q , $P \cdot Q$ in Q , $Q \cdot Q$ in P and $P \cdot P$ in Q give 2, 2, 1, 8, $\frac{1}{2} \neq 4$, so (V) fails. The reading $P \cdot Q$ (or $Q \cdot P$) in P gives 4, which violates (B) since $4 > 2$. $\square$

Remark 4. Table 1 shows that no single k makes the value clause alone fail under *all* readings. For example, the non-standard reading $P \cdot Q$ in P gives 4 for every $k \geq 2$, which equals 2^{k-1} at $k = 3$. But the two clauses of the conjecture are incompatible for every $k \geq 3$, whatever M means: (B) gives $M((k), (k)) \leq 2$, while (V) demands $M((k), (k)) = 2^{k-1} \geq 4$. Part (2) of Theorem 3 makes this concrete with the actual coefficients.

4 The case $k = 3$ in three variables

This section gives the finite computation formalised in Lean. It proves part (1) and part (2) of Theorem 3 at $k = 3$ without using the basis theorem.

The strict partitions of 6 are 6, 51, 42, 321, all of length ≤ 3 . Specialise to three variables x_1, x_2, x_3 . Let m_1, m_2, m_3, m_4 be the monomials $x_1^6, x_1^5 x_2, x_1^4 x_2^2, x_1^3 x_2^2 x_3$. Their coefficients are:

	m_1	m_2	m_3	m_4
Q_6	2	4	4	8
Q_{51}	0	4	8	24
Q_{42}	0	0	4	24
Q_{321}	0	0	0	8
Q_3Q_3	4	16	32	112
P_3Q_3	2	8	16	56
P_3P_3	1	4	8	28

For instance $Q_{321}(x_1, x_2, x_3) = 8x_1x_2x_3(x_1 + x_2)(x_1 + x_3)(x_2 + x_3)$. The 4×4 block is triangular with nonzero diagonal, so $Q_6, Q_{51}, Q_{42}, Q_{321}$ are linearly independent in $\mathbb{Q}[x_1, x_2, x_3]$. Suppose $Q_3Q_3 = \sum_{\nu} c_{\nu}Q_{\nu}$ in $\Gamma_{\mathbb{Q}}$ (or any rational identity of this shape). Specialising to three variables gives the same identity there. Comparing coefficients at $m_1, \dots, m_4$ gives $2c_6 = 4$, $4c_6 + 4c_{51} = 16$, $4c_6 + 8c_{51} + 4c_{42} = 32$ and $8c_6 + 24c_{51} + 24c_{42} + 8c_{321} = 112$. So $(c_6, c_{51}, c_{42}, c_{321}) = (2, 2, 2, 0)$, and the coefficients are forced. Existence of the expansion in Γ is Theorem 1 (proved in Γ); the three-variable computation gives uniqueness. Together these determine $M((3), (3)) = 2$ in the Q -reading. The same argument applies to the other seven readings, using $P_{\nu} = 2^{-\ell(\nu)}Q_{\nu}$. The forced vectors are $(1, 2, 2, 0)$ for P_3P_3 in P , $(2, 4, 4, 0)$ for P_3Q_3 in P , $(1, 1, 1, 0)$ for P_3Q_3 in Q , $(4, 8, 8, 0)$ for Q_3Q_3 in P , and $\frac{1}{2}(1, 1, 1, 0)$ for P_3P_3 in Q .

5 Lean formalisation

The directory `lean4/` is a Lean 4.19.0 project that uses only the core library. It contains no `sorry`, `native_decide` or added axioms; heavy closed computations use `decide +kernel`.

- *Polynomials.* A polynomial in $\mathbb{Z}[x_1, x_2, x_3]$ is a list of (exponent triple, coefficient) pairs. `coeff p m` sums all coefficients with exponent `m`. `add`, `scale` and `mul` are proved to act on `coeff` as expected (`coeff_add`, `coeff_scale`).
- *Objects.* `q r` is the coefficient of t^r in $\prod_{i \leq 3} (1 + x_i t) / (1 - x_i t)$, built monomial by monomial. `Q2 r s` is (2). `Qfun` is the Pfaffian (3), and `Pfun` $\nu = 2^{-\ell(\nu)}Qfun \nu$. `P_exact` proves that this division is exact. `strictParts` enumerates strict partitions, and `basis Z n` is $\{Z_{\nu} : \nu \text{ strict}, |\nu| = n\}$. As sanity checks, `q_relation` proves (1) for $r \leq 6$ and `Q321_eq` proves the product formula for Q_{321} .
- *The clause.* `Expands d T cs B` says that $d \cdot T = \sum_i c_i B_i$ coefficientwise. With $d > 0$ this is a rational expansion with coefficients c_i/d . `ValueAt k X Y Z` says that some expansion of $X_{(k)}Y_{(k)}$ in the Z -basis has largest coefficient 2^{k-1} . `BoundAt k X Y Z` says that every such expansion has largest coefficient $\leq 2^{\ell((k))}$.
- *Results.* `QQ_in_Q`, `PP_in_P` and six more theorems prove the expansions as polynomial identities. `expansion_exists` shows non-vacuity, and `solveQ`, `solveP` and `expansion_unique` prove uniqueness by triangularity. `max_coeff` computes $M((3), (3))$ for each reading. `value_clause_false_P` and `value_clause_false_Q` are $\neg \forall k \geq 1, ValueAt k$ in the two standard readings. The main theorem `conjecture_00000001196.false` states $\forall X Y Z, \neg(ValueAt 3 X Y Z \wedge BoundAt 3 X Y Z)$. It is assembled from `value_fails` and `bound_fails`. `bound_holds_P` and `value_holds_PQP` show that neither predicate is trivially false.

Scope. Lean works in three variables. The $k = 3$ instance is faithful to the statement in Γ because the three-variable expansion is unique there: all strict partitions of 6 have at most three parts, specialisation to three variables is a ring homomorphism that commutes with all constructions, and the four basis functions are proved linearly independent (Section 4). Hence `ValueAt 3` and `BoundAt 3` say exactly what the conjecture says at $\lambda = \mu = (3)$, and the main theorem is stated only for this instance. The general- k statement (Theorem 1 and Table 1) and the existence of the expansion in Γ are proved in this report rather than in Lean.

6 Independent Python check

`verify.py` uses only the Python standard library and a different method from Lean. It computes Q_ν and P_ν by enumerating marked shifted tableaux. It checks that $q_r = \sum_{a+b=r} e_a h_b = Q_{(r)}$ and the relation (1). It checks that the Pfaffian formulas used in Lean agree with the tableau definition for all strict $\nu \vdash 6$. It also checks that $Q_\nu = 2^{\ell(\nu)} P_\nu$. For $k = 1, \dots, 5$, with as many variables as the longest strict partition of $2k$ needs, it expands all products in both bases by exact rational Gaussian elimination, asserting full rank. It reproduces Table 1 and confirms Theorem 1 and Corollary 2 for $k \leq 5$. It also prints the coefficient table of Section 4.

References

- [1] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., Oxford University Press, 1995, Chapter III, Section 8.
- [2] J. R. Stembridge, Shifted tableaux and the projective representations of symmetric groups, *Adv. Math.* 74 (1989), 87–134.